

SIERRA BLANCA, NM, USA

WMO: 722683

Lat: 33.467N

Lon: 105.533W

Elev: 6810

StdP: 11.42

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 93083

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.4	18.7	-8.8	4.4	38.8	-4.1	5.7	40.3	40.7	44.1	34.8	45.7	6.7	140	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	23.6	87.9	55.9	84.4	55.8	82.2	56.1	64.2	72.2	62.9	71.8	61.7	71.6	11.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	110.7	66.8	60.8	103.1	65.4	58.8	95.6	64.6	33.5	71.6	32.1	71.5	31.3	73.0	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.9	26.4	22.5	7.4	92.1	6.2	1.9	3.0	93.5	-0.7	94.6	-4.2	95.6	-8.7	97.0		
(5)			6.3	65.7	5.4	2.0	2.4	67.2	-0.8	68.4	-3.8	69.5	-7.7	71.0		(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	38.7	40.9	46.6	52.7	61.2	69.5	70.3	68.2	63.7	54.8	45.6	38.2		
	DBStd	13.72	8.27	9.18	8.13	7.91	7.35	5.10	4.01	3.97	5.24	7.59	8.54	8.47		
	HDD50	1452	355	271	160	65	10	0	0	0	1	42	178	370		
	HDD65	4443	815	675	570	370	157	17	6	15	83	322	582	831		
	CDD50	3009	5	16	55	146	357	585	630	564	412	189	46	4		
	CDD65	527	0	0	0	1	40	152	172	114	44	4	0	0		
	CDH74	4756	0	0	1	24	470	1664	1383	798	372	44	0	0		
	CDH80	1224	0	0	0	1	94	544	380	164	39	2	0	0		
(14) Wind	WSAvg	9.0	9.3	10.7	10.8	12.1	10.6	8.9	6.8	6.0	6.8	8.3	8.6	9.5		
(15) Precipitation	PrecAvg	16.4	0.7	0.7	0.7	0.7	1.1	1.2	3.5	2.9	2.1	1.2	0.6	0.9		
	PrecMax	22.0	1.7	2.3	1.9	3.0	4.0	4.0	7.2	7.4	4.2	4.0	1.7	3.1		
	PrecMin	8.3	0.0	0.0	0.1	0.1	0.1	0.1	1.1	0.7	0.4	0.0	0.0	0.0		
	PrecStd	3.1	0.5	0.5	0.4	0.7	1.1	1.1	1.4	1.4	1.0	0.9	0.5	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.2	66.2	72.3	77.1	86.3	91.0	90.3	87.8	82.5	79.0	70.5	63.1		
		MCWB	42.5	44.4	46.0	48.8	54.3	56.2	57.5	57.7	56.7	52.6	46.9	41.7		
	2%	DB	57.4	63.3	68.1	73.1	82.2	88.4	86.3	83.8	81.2	74.6	66.4	57.4		
		MCWB	39.7	42.3	44.5	46.7	52.0	54.9	57.6	58.0	56.3	51.7	45.4	40.0		
	5%	DB	54.6	59.2	64.5	71.6	79.3	85.8	83.9	81.5	78.9	72.1	63.6	54.7		
		MCWB	38.7	40.2	42.6	46.3	50.4	54.3	57.8	59.0	55.7	50.7	44.1	38.6		
	10%	DB	51.9	55.1	62.7	67.8	75.4	82.4	81.6	79.0	75.2	68.0	59.1	51.6		
		MCWB	37.3	38.3	41.9	44.6	48.9	53.9	58.4	59.2	55.6	49.5	42.1	37.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	46.5	49.4	54.2	59.3	63.1	65.9	66.5	64.1	60.3	50.7	45.2		
		MCDB	56.8	60.4	64.9	65.4	68.5	76.0	73.4	73.3	70.9	66.2	62.5	55.3		
	2%	WB	42.4	44.0	46.4	50.4	56.2	60.9	64.0	64.6	61.9	57.0	47.7	42.0		
		MCDB	53.8	58.4	62.1	65.8	71.1	72.9	72.9	71.7	69.6	66.1	60.2	54.8		
	5%	WB	40.1	41.9	44.3	48.1	54.3	59.5	62.6	63.2	60.5	54.4	45.8	40.0		
		MCDB	52.1	56.4	60.8	64.5	70.6	73.4	72.8	71.0	69.1	64.8	58.6	52.2		
	10%	WB	38.1	39.7	42.3	46.1	52.4	58.1	61.5	61.9	59.1	52.2	43.9	38.0		
		MCDB	49.5	53.9	58.9	63.6	69.3	72.6	72.6	71.1	68.6	63.5	57.6	49.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	20.7	22.6	23.2	22.9	23.6	21.3	19.9	20.4	21.2	20.9	19.3		
		MCDBR	22.3	24.0	25.5	25.7	25.9	26.5	24.7	22.9	23.7	23.9	23.8	22.0		
	5% WB	MCWBR	12.2	12.3	12.6	12.3	11.4	10.8	8.6	8.3	9.1	11.1	12.3	11.9		
		MCWBR	20.6	22.0	24.0	24.7	23.4	22.7	20.1	18.6	19.1	19.9	21.6	20.4		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.233	0.250	0.267	0.284	0.309	0.341	0.324	0.294	0.263	0.242	0.231		
	taud		2.704	2.690	2.627	2.565	2.569	2.509	2.475	2.503	2.569	2.657	2.655	2.682		
	Ebn at Noon		325	331	330	325	317	307	297	302	309	315	315	318		
	Edh at Noon		22	24	28	32	32	34	35	33	30	25	23	21		
(44) All-Sky Solar Radiation	RadAvg		1114	1426	1831	2247	2412	2370	2099	1941	1784	1489	1180	989		
	RadStd		81	120	108	120	122	167	133	110	103	137	85	60		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	-3.49	N/A	-1.22	-2.16	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page